Multi-Source Weather Dashboard Web-Application |Documentation and Report

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21. October 2026

1 Project Overview & Architecture

The goal of this project is to develop a robust, responsive weather dashboard having strict native web standards: HTML5, CSS3, and vanilla JavaScript.

2 Directory Structure

I tried as much as I could to make the presentation, operational logic, and styling clean and maintainable. `-index.html`: has a good structural layout and semantic DOM foundation.

- `css/style.css`: It helps give the site a responsive design system.
- `js/api.js`:helps in Multi-source data fetching and data normalization.
- `js/app.js`:is responsible for lifecycle controller application, the DOM rendering, and Chart.js integration.
- `docs/documentation.pdf`: my project report and AI declaration.

3 Tools & External Services

- **Vanilla JavaScript (ES6+):** Native Fetch API, asynchronous flow (`async`/`await`), and `localStorage` for client-side state persistence.
- **Modern CSS3:** offers a Fluid layout architecture built with Flexbox, CSS Grid, and custom properties (variables) for smooth theme switching.
- **Chart.js (via CDN):** responsive for a Lightweight HTML5 `<canvas>` rendering engine used to plot simultaneous multi-provider forecast curves.
- **Data Providers & APIs:**
  - **Open-Meteo API:** is a Primary engine providing geocoding, current conditions, 24-hour hourly slices, and 7-day forecasts.
  - **OpenWeatherMap API:** is a Secondary data source for temperature verification.
  - **WeatherAPI :** is a Tertiary independent data source fulfilling multi-source criteria.
  - **Browser Geolocation API :** offers hardware-level GPS coordinate retrieval.

4 User Manual

- **Location Search:** Use the search bar to enter any city name and press **Search** to trigger geocoding and update all dashboard views.
- **GPS Detection:** Click the live location icon to immediately fetch weather data for your current location.
- **Temperature Scale Toggle:** Click **°C**, **°F**, or **K** in the top toolbar to convert all values across the current card, hourly rail, and weekly forecast.

- **Comparative Forecast:** Inspect the **24-Hour Forecast Comparison** chart to analyze concurrent temperature trends plotted from separate data providers.
- **Favorites Management:** Click the favorite button to store the current city in your browser's storage.

5 Points Justification (Target: 40 / 40 Points)

Feature / Criteria	Max	Claim	Implementation Details
Well-written PDF report	3	3	Comprehensive 2-page report with architecture, manual, rubric justification, and AI declaration.
Responsive application	4	4	Fluid CSS Grid & Flexbox architecture functioning across mobile, tablet, and desktop screens.
Cross-browser compatibility	3	3	Validated across Chrome, Firefox, Safari, and Microsoft Edge using standard Web APIs.
Clear directory structure	2	2	Strict separation of markup (`index.html`), styling (`css/`), scripts (`js/`), and documentation (`docs/`).
Search for locations	1	1	Integrated search input utilizing Open-Meteo's geocoding endpoint for coordinates.
GPS Geolocation API	2	2	One-click geolocation fetching local device coordinates via `navigator.geolocation`.
At least two data providers	3	3	Aggregates independent forecast feeds from Open-Meteo and OpenWeatherMap.
At least three data providers	2	2	Integrates WeatherAPI as an independent third weather data source.
Current weather display	1	1	Real-time temperature, condition summary, and weather icons.
24-hour hourly forecast	3	3	Horizontally scrollable hourly cards displaying temperatures and conditions for the next 24 hours.
7-day forecast	3	3	Extended weekly outlook detailing daily high/low temperatures and weather icons.
Weather forecast icons	3	3	Contextual condition icons rendered alongside numeric data across all forecast components.
Dynamic weather styling	2	2	Background and accent colors adapt dynamically to temperature (cold/hot) and solar cycle (day/night).
Dual-source graph	3	3	Interactive Chart.js canvas plotting simultaneous comparative forecast lines.
Favorites menu	2	2	Allows bookmarking locations with persistence across browser sessions via `localStorage`.
Unit conversion (°C/°F/K)	2	2	Instant mathematical conversions between Celsius, Fahrenheit, and Kelvin across all views.
Custom feature: Solar state engine	1	1	Real-time day/night contrast engine adapting typography and card backgrounds to ambient conditions.
Total Score Claimed	**40**	**40**	**Full fulfillment of all project criteria and general course requirements.**

6 Mandatory Declaration of AI Usage

In accordance with course submission guidelines:

1. **AI Systems Utilized:**
   - Gemini (Google)
2. **Nature and Scope of AI Assistance:**
   - **Project Architecture Review:** AI was used at the beginning during planning so as to be certain of following the project criteria.
   - **Technical Troubleshooting:** AI was also used to correct wrongly written code and suggest better versions and also to debug unit conversion formatting.
   - **Documentation Structuring:** It also served as a tool for drafting and proof reading so that sentences can be smooth and the report in Typst markup should look better.
3. **Verification & Authorship:**
   - All source code, API integrations, styles, and final repository commits were tested, validated, and assembled independently by the student.